

Effect of Cadence on Physiological and Perceptual Responses during Eccentric Cycling at Different Power Outputs

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ABSTRACT

MATER, A., A. BOLY, H. ASSADI, A. MARTIN, and R. LEPERS. Effect of Cadence on Physiological and Perceptual Responses during Eccentric Cycling at Different Power Outputs. *Med. Sci. Sports Exerc.*, Vol. 55, No. 6, pp. 1105–1113, 2023. **Introduction:** The effect of cadence in eccentric (ECC) cycling on physiological and perceptual responses is, to date, poorly understood. This study aimed to evaluate the effect of cadence during ECC cycling on muscular activation (EMG), oxygen consumption ($\dot{V}O_2$), and perceived effort (PE) for two different levels of power output. **Methods:** Seventeen participants completed four sessions 1 wk apart: 1) determination of the maximal concentric peak power output (PPO) and familiarization with ECC cycling at five cadences (30, 45, 60, 75, and 90 rpm); 2) second familiarization with ECC cycling; 3) and 4) ECC cycling exercise consisting of 5 min at the five different cadences at either 40% or 60% PPO. PE was reported, and $\dot{V}O_2$ and EMG of seven muscles were calculated over the exercise's last minute. **Results:** PE, $\dot{V}O_2$, and global lower limb muscles activation (EMG_{ALL}) showed an effect of cadence ($P < 0.001$) and followed a curvilinear function. Both low and high cadences increased PE and $\dot{V}O_2$ responses compared with intermediate cadences. Although muscle activation of vastus lateralis follows a U-shaped curve with cadence, it was greater at low cadence for rectus femoris and biceps femoris, greater at high cadence for tibialis anterior and gastrocnemius medialis, and was not altered for soleus. The estimated optimal cadence was greater (all $P < 0.01$) for $\dot{V}O_2$ (64.5 ± 7.9 rpm) than PE (61.7 ± 9.4 rpm) and EMG_{ALL} (55.9 ± 9.3 rpm), but power output had no effect on the optimal cadences. **Conclusions:** The physiological and perceptual responses to changes in cadence during ECC cycling followed a U-shaped curve with an optimal cadence depending on the parameter considered. **Key Words:** ECCENTRIC CONTRACTION, CYCLING CADENCE, MUSCULAR ACTIVATION, PERCEIVED EFFORT, OXYGEN CONSUMPTION

Since the first work of Abbott et al. (1) on eccentric (ECC) cycling evidencing lower cardiorespiratory stress compared with concentric cycling, more and more studies have focused on this unconventional form of exercise. ECC cycling consists of resisting the pedals driven backward by a motor, so knee and hip extensors absorb work through ECC contraction (i.e., active lengthening of a contracting muscle) (2). Maximal ECC muscle action allows greater absolute force production than maximal concentric action (i.e., shortening muscle action) (3–5). This higher force production capacity associated with lesser energy expenditure (6,7) suggests that molecular mechanisms responsible for such mechanical behavior involve contractile and noncontractile proteins (6,7). The main benefit of this contraction mode is that, compared with the concentric mode performed at the same mechanical work rate (or power output during cycling), the ECC mode results in a

lower metabolic cost (8), cardiorespiratory stress (1,9), perceived effort (PE) (10), and muscle activation (11). Moreover, because repetitive high muscle contractions are also a stimulus to promote strength and muscle mass gains (12), ECC cycling is a valuable exercise for rehabilitating patients suffering, for example, from cardiorespiratory diseases (13).

The study of cadence has been a historical focus in conventional cycling (i.e., concentric cycling). When cadence changes, physiological and perceptual parameters (e.g., oxygen consumption [$\dot{V}O_2$], muscular activation, and PE) follow a curvilinear function with a minimum value reached at a pedaling cadence generally referred to as “optimal cadence” (i.e., a cadence that minimizes the considered parameter). In other words, increasing or decreasing the cadence from the optimal cadence increases physiological and perceptual responses (14–16). However, although much research has been conducted to determine the optimal cycling cadence, exercise characteristics (i.e., duration, intensity, and slope) and dependent variables of interest make the task difficult (17). Indeed, there is no single optimal cadence, but as many as there are physiological and perceptual parameters of interest. It has been suggested that perceptive response is a compromise between energetical optimal cadence (i.e., lowest $\dot{V}O_2$), which appears at relatively low cadences, and neuromuscular optimal cadence (i.e., lowest muscle activation) observed at high cadences (18). Furthermore, optimal cadences tend to increase with power output (19–21), regardless of the parameters considered.

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Although the effect of cadence on physiological response in concentric cycling is widely accepted, studies of ECC cycling typically set cadence arbitrarily between 15 and 60 rpm (10,22), and only a few studies have explored the physiological effect of different cadences during submaximal ECC cycling (23–26). For example, Wells et al. (23) compared three cadences (30, 60, and 90 rpm) at four power outputs (“zero load,” 60, 120, and 180 W) and found that metabolic cost was higher at 30 rpm than at 60 and 90 rpm. Conversely, Chung et al. (24) found that oxygen consumption and heart rate were greater at 75 than 55 rpm (and not 35 rpm) for a power output of 60 W. Moreover, in both studies of Knuttgen et al. (25,26), oxygen consumption was greater at 100 and 20 rpm compared with 60 rpm for different power outputs. Nevertheless, it was difficult to assess the physiological response with only three cadences and a small number of participants (three and two in [23], and [25,26], respectively). Moreover, previous studies have focused only on cardiorespiratory function but have not examined neuromuscular aspects or perceptual responses relevant to understanding ECC cycling and the field of application of rehabilitation.

In this context, this study aimed to investigate the effect of cadence in ECC cycling, considering oxygen consumption, muscle activation, and PE at two different power outputs. We expected that (i) physiological and perceptual responses to pedaling cadence would follow curvilinear curves with different optimal cadences depending on the parameter considered, and (ii) optimal cadence would increase with power output.

METHODS

Participants. A required sample size of 14 was calculated based on cardiorespiratory responses to cadence between 55 and 75 rpm from Chung et al. (24); for a repeated-measures between-factors test, an effect size of $d = 0.61$ —calculated from presented data—an alpha level of 0.05, and a power 0.80. Then 17 healthy volunteers (12 males, 5 females) who provided written informed consent participated in this study (age = 22.2 ± 1.6 yr, height = 176.2 ± 9.7 cm, body mass = 69.7 ± 9.9 kg). The experimental design of the study was approved by the regional ethics committee (CPP EST: approval no. A00064-49). Most participants were novices to ECC cycling and only practiced concentric cycling for daily commutes. Only three took part in an ECC cycling experiment 6 months before the present one. The participants were free from injuries in their lower limbs during the last 3 months. Their mean concentric cycling peak power output (PPO) was 275.0 ± 57.7 W, corresponding to a peak oxygen uptake ($\dot{V}O_{2\text{peak}}$) of 46.5 ± 10.0 mL·kg⁻¹·min⁻¹ and a maximal heart rate of 183.6 ± 8.0 bpm. Then they were defined as recreationally trained (27,28).

Experimental procedure. Each participant performed two familiarizations and two experimental sessions in the laboratory at a similar time of the day (± 2 h), separated by 1 or 2 wk. Participants were asked to abstain from fatiguing exercise the day before each session and maintain their usual diet.

The first familiarization session started with a concentric incremental ramp test, starting at 50 W with a linear increase of 1 W/3 s until participants could not produce power output anymore (i.e., when pedaling cadence dropped below 50 rpm for more than 5 s). This test allowed us to determine PPO, $\dot{V}O_{2\text{peak}}$, and maximal heart rate. After a 15-min recovery period, participants had to match different cadences (30, 45, 60, 75, and 90 rpm) for five bouts of 3 min of ECC cycling (at 60% concentric peak power output), separated by 2 min of recovery. The purpose of this session was to familiarize the participants with the different cadences, the PE scale (CR100 scale of Borg), and the procedure of the squat jump test used as an indirect index of fatigability (11,29). The second familiarization session was constructed similarly to the experimental sessions, allowing participants to be familiar with the complete protocol. Both sessions produced a repeated bout effect, reducing muscle disruptions led by unaccustomed ECC exercise (30). In this sense, none of them reported delayed muscle soreness after experimental sessions.

The subsequent two sessions were started with a 10-min warm-up in concentric cycling at 30% PPO, followed by 1 min at PPO, during which EMG was recorded. Then participants performed the five bouts of 5 min of ECC cycling at the different cadences with 3 min of recovery, preceded and followed by the assessment of squat jump height. These third and fourth sessions were used to evaluate the effect of the different cadences at 40% and 60% PPO during ECC cycling on physiological and perceptual parameters. Both power outputs (for experimental sessions) and cadences (for each session) were completed in a randomized order. Experimental sessions were separated by at least 1 wk to a maximum of 2 wk.

Cycling exercise. A concentric semirecumbent cycle ergometer (Ergoline GmbH, Ergoselect 600, Germany) was used for the incremental test and warm-up, and an ECC cycle ergometer (Cyclus 2, Cyclus GmbH, Germany) was used for all ECC cycling exercises. Positions between cycles were adjusted with the seat seatback and incline to match the knee and trunk angles as much as possible. The cycle ergometer settings were the same between familiarization and experimental sessions. The ECC cycle ergometer was also set on isopower mode, meaning that participants had to match an asked pedaling cadence for given power output.

Respiratory gas exchanges, heart rate, and muscle activation (EMG) were assessed during the last minute of the 5-min ECC cycling. The participants' perception of effort was asked during the same period. We assessed squat jump height before and immediately after pedaling exercises. The time latency between the end of the last bout of cycling exercise and the onset of the jump test was 1 min.

Electromyography. Bipolar wireless EMG units (Pico Cometa, Biometrics, France) were fixed on Ag/AgCl electrodes (recording over 10 mm; a center-to-center distance of 15 mm). The skin of the participants was shaved and cleaned with alcohol swabs. Electrodes were positioned on the belly of the vastus lateralis (VL), rectus femoris (RF), biceps femoris (BF), gastrocnemius medialis (GM), soleus (SOL), tibialis anterior (TA), and

biceps brachii (BB) muscles, following the recommendations of Barbero et al. (31). BB muscle was assessed because power output seems to affect postural trunk stabilization (32) and preliminary experimentation let us think that cadence could also lead to a different level of tension in the upper body. At the beginning of each session, electrodes were carefully replaced using anatomical landmarks. The signal was assessed at a frequency of 2000 Hz, amplified 1000 times, stored for offline analysis, and then band-pass filtered between 10 and 390 Hz.

Cardiorespiratory responses. Heart rate and oxygen consumption ($\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) were assessed with a chest strap heart rate monitor (model V800; Polar Electro Oy, Kempele, Finland) and metabolic analyzer K5 (COSMED, Rome, Italy), respectively. The gas analyzer was calibrated and used according to company guidelines in a closed room with constant temperature and humidity.

Perceptual response. The perception of effort (i.e., difficulty breathing and driving legs to pedal) (33) was asked to the participants during the last minute of each bout. The perception exertion was reported using Borg's Cr100 scale (34). The two familiarization sessions allow participants to familiarize themselves with this scale.

Data analysis. Electromyography and oxygen consumption data were considered outliers and excluded if they were 2.5 SD below and above the mean values. Data from oxygen consumption of two sessions were excluded because of acquisition trouble with the metabolic analyzer. The total of removed data represented 0.73%. Heart rate data were averaged from 3 min 45 s to 4 min 45 s of each bout. In the same way, oxygen consumption data were averaged over this range of time to obtain a 1-min steady-state plateau. During this lap time, the EMG of each muscle was quantified by the root-mean-square value and normalized pedaling to the EMG root-mean-square obtained during the pedaling minute at PPO performed at the end of the warm-up of the same session. The lower limb muscle activation was first analyzed separately for each muscle. Second, it was averaged (EMG_{ALL}) to define the lower limb muscle activation behavior according to power output and cadences. Oxygen consumption and heart rate during exercise were expressed in percentage of the $\dot{V}\text{O}_{2\text{peak}}$ and maximal heart rate at the end of the incremental test. The value of oxygen consumption was averaged at the last minute of the plateau, and we admitted the steady state because the cardiorespiratory responses were likely under the anaerobic threshold (35). Minute ventilation, respiratory frequency, and respiratory quotient were expressed in liter per minute, breath by minute, and arbitrary unit, respectively.

For each participant, optimal cadences were estimated for $\dot{V}\text{O}_2$, EMG_{ALL} , and PE at both power outputs. We first determined an n th-order polynomial using the least-squares method with cadence as the independent variable and $\dot{V}\text{O}_2$, EMG_{ALL} , and PE as the dependent variable. Optimal cadences followed a second- or third-order polynomial regression. Therefore, the polynomial regression model that maximized the data fit was deemed appropriate for subsequent analysis. Then when data followed a quadratic regression, the optimum was calculated

as the minimum of the function. We used cubic regression if the estimated optimal cadence was below or above the range from 30 to 90 rpm. For this latter, the optimum was calculated by finding the derivative of the third-order regression for which the second derivative was positive. Then the corresponding cadence was determined as the estimated optimal cadence for each participant, power output, and parameter considered. Therefore, six optimal cadences were calculated for each participant. The correlation coefficient for each parameter, and each participant, was determined. Then they were presented with the resulting equation that described the effect of cadence on our parameters of interest as mean and the range of minimal and maximal individual data. The polynomial equation fit most of the individual data well: r values were 25% below 0.9, 56% between 0.9 and 0.95, and 19% above 0.95. Presented polynomial equations showed the average of the individual regressions.

Statistical analysis. Statistical analysis was performed with the software JAPS (version 0.16.1; JASP TEAM, Amsterdam, Netherlands). According to our hypotheses, we compared the effect of cadences and power output on different physiological and perceptual parameters. Therefore, we used two-way repeated-measures ANOVA (power output–cadence). Moreover, estimated optimal cadences for those parameters were compared between them, and their evolution across power outputs using two-way repeated-measures ANOVA (power output–parameters). ANOVA was used when sphericity (Mauchly's test) was verified. When significance was reached on at least one of the factors of ANOVA, the Holm test was used for *post hoc* analysis. A Greenhouse–Geisser correction was applied if the sphericity of the data was not respected.

RESULTS

Jump test. Squat jump height (28.15 ± 7.16 cm) did not show power output–time interaction ($P = 0.546$, $\eta_p^2 = 0.023$), power output ($P = 0.177$, $\eta_p^2 = 0.111$), or time ($P = 0.062$, $\eta_p^2 = 0.201$) effect, suggesting that fatigability was minimal after ECC cycling exercises.

Cardiorespiratory responses. Table 1 details the heart rate responses and the ventilation parameters to the changes in cadence and power. ANOVA did not show power output–cadence interaction effect for all parameters except heart rate ($P < 0.05$, $\eta_p^2 = 0.167$). However, at 40% PPO, the heart rate was lower at 45 and 60 rpm than at 30 and 90 rpm; at 60% PPO, it was greater at 30 rpm than at all other cadences. Respiratory frequency ($P < 0.05$, $\eta_p^2 = 0.331$) and minute ventilation ($P < 0.001$, $\eta_p^2 = 0.559$) showed a main power output effect. Modifying cadence altered respiratory frequency ($P < 0.05$, $\eta_p^2 = 0.296$) and minute ventilation ($P < 0.05$, $\eta_p^2 = 0.265$). Respiratory frequency was greater at 90 rpm than at 30, 45, and 60 rpm, and minute ventilation was greater at 30 and 90 rpm compared with 60 rpm. The respiratory exchange ratio did not show a main power output ($P = 0.542$, $\eta_p^2 = 0.027$) nor cadence ($P = 0.075$, $\eta_p^2 = 0.158$) effect.

TABLE 1. Cardiorespiratory responses to different cadences.

		Power (% PPO)		Cadence (rpm)						
				30	45	60	75	90		
Heart rate (% max)	40	£	53 ± 5	51 ± 5	<i>a e</i>	51 ± 6	<i>aa ee</i>	52 ± 5	53 ± 5	
	60		60 ± 7	57 ± 7	<i>aaa</i>	56 ± 8	<i>aaa</i>	56 ± 7	<i>aaa</i>	<i>aa</i>
Respiratory frequency (bpm)	400	\$#	24 ± 3	23 ± 2	<i>ee</i>	23 ± 3	<i>ee</i>	24 ± 3	25 ± 4	
	60		25 ± 3	25 ± 3		25 ± 3		27 ± 4	27 ± 6	
Minute ventilation (L·min ⁻¹)	40	\$\$\$#	17.1 ± 4.1	15.9 ± 4.0		15.4 ± 4.0	<i>a e</i>	16.1 ± 3.4	18.4 ± 3.3	
	60		23.3 ± 6.3	20.3 ± 5.2		19.6 ± 5.0		20.3 ± 7.1	22.5 ± 9.7	
Respiratory quotient (a.u.)	40		0.75 ± 0.05	0.75 ± 0.05		0.75 ± 0.06		0.76 ± 0.05	0.76 ± 0.07	
	60		0.78 ± 0.07	0.75 ± 0.06		0.76 ± 0.08		0.77 ± 0.05	0.78 ± 0.07	

The main cadence effect was indicated above, below, and in the middle for 40% PPO, 60% PPO, and noninteraction effect, respectively.

\$, main power output effect; #, main cadence effect; £, power output–cadence effect interaction; *a*, different from the condition at 30 rpm; *b*, different from the condition at 45 rpm; *c*, different from the condition at 60 rpm; *d*, different from the condition at 75 rpm; *e*, different from the condition at 90 rpm. One sign, $P < 0.05$; two signs, $P < 0.01$; three signs, $P < 0.001$.

Oxygen consumption showed power output–cadence effect ($P = 0.034$, $\eta_p^2 = 0.196$) (Fig. 1A). At 40% PPO, the metabolic response was greater at 30 and 90 rpm than at 75 rpm ($P < 0.05$, $df > 0.470$). At 60% PPO, it was greater at 30 rpm than at all other cadences (all $P < 0.05$, $df > 0.488$).

Muscular activation. EMG_{ALL} for the different cadences and power outputs is depicted in Figure 1C. EMG_{ALL} ($P < 0.05$, $\eta_p^2 = 0.183$) showed a significant power output–cadence effect. However, at 40% PPO, 90 rpm elicited greater muscular activation than 45, 60, and 75 rpm; at 60% PPO, it was greater at 30 rpm than 45 rpm, and 90 rpm caused greater EMG than at 45, 60, and 75 rpm.

EMG activities of the lower limb muscles are presented in Figure 2. Considering muscles distinctly, only RF ($P < 0.05$, $\eta_p^2 = 0.178$) and GM ($P < 0.01$, $\eta_p^2 = 0.274$) showed a power output–cadence interaction effect, whereas all others did not (all $P > 0.166$, all $\eta_p^2 < 0.105$). Muscular activation of VL ($P < 0.01$, $\eta_p^2 = 0.490$), RF ($P < 0.05$, $\eta_p^2 = 0.316$), and TA ($P < 0.05$, $\eta_p^2 = 0.330$) muscles were greater at 60% PPO compared with 40% PPO, whereas GM and SOL did not (all $P > 0.075$, all $\eta_p^2 < 0.185$). Then a cadence effect was found for VL, BF, and TA muscles (all $P < 0.022$, all $\eta_p^2 > 0.192$) but not SOL ($P = 0.177$, $\eta_p^2 = 0.108$).

Regarding the upper limb, BB muscle showed a power output–cadence interaction effect ($P < 0.05$, $\eta_p^2 = 0.191$) (Fig. 3). *Post hoc* analysis reveals no effect of cadence at 40% PPO. At 60% PPO, BB muscle activation was greater at 30 rpm than all other cadences (all $P < 0.05$, $df > 0.527$).

Perception of effort. PE (Fig. 1B) showed both power output ($P < 0.01$, $\eta_p^2 = 0.368$) and cadence ($P < 0.001$, $\eta_p^2 = 0.416$) main effect but not power output–cadence interaction effect ($P = 0.091$, $\eta_p^2 = 0.133$). Perceived exertion was greater at 60% PPO compared with 40% PPO. The perceived response was greater at 30 rpm compared with 45, 60, and 75 rpm (all, $P < 0.001$, $df > 0.654$) and at 90 rpm compared with 60 rpm ($P < 0.001$, $df = 0.491$).

Optimal cadences. Physiological and perceptual responses to cadence follow a U-shaped curve. The effect of cadence could be appreciated through the following functions:

- $8.55 \cdot 10^{-5}x^2 - 9.92 \cdot 10^{-3}x + 6.22 \cdot 10^{-1}$ with r between 0.32 and 0.99 for EMG_{ALL},
- $4.44 \cdot 10^{-3}x^2 - 5.73 \cdot 10^{-1}x + 4.09 \cdot 10^1$ with r between 0.88 and 0.99 for $\dot{V}O_2$,

- $4.40 \cdot 10^{-3}x^2 - 5.09 \cdot 10^{-1}x + 4.54 \cdot 10^1$ with r between 0.39 and 0.99 for PE.

Although ANOVA revealed no effect of power output ($P = 0.779$, $\eta_p^2 = 0.006$) on estimated optimal cadences (Fig. 4), this was later different between parameters ($P = 0.006$, $\eta_p^2 = 0.318$). *Post hoc* analysis showed that the optimal cadence of oxygen consumption (64.5 ± 7.9 rpm) was greater than optimal from PE (61.7 ± 9.4 rpm, $P < 0.01$, $df > 0.904$) and EMG_{ALL} (55.9 ± 9.3 rpm, $P < 0.001$, $df > 0.491$).

DISCUSSION

The present study is the first to describe both physiological and perceptual responses to changes in cadence for ECC cycling. We hypothesized that physiological and perceptual responses would adopt a curvilinear function with different optimal cadences. We also expected that the optimal cadences would increase with power output. The results showed that physiological and perceptual responses followed individual U-shaped curves, but the estimated optimal cadences were unaffected by power output.

All participants performed a jump test before and after pedaling the ECC cycling exercise to verify that the experimentation was conducted without inducing muscle fatigability. As the subjects were able to maintain all imposed cadences throughout each session without altering squat jump performance, familiarization and protocol's design limited the effect of fatigue on muscular function. The present familiarization likely had a protective effect on subsequent ECC sessions (30).

Although ECC cycling is now used in rehabilitation, no study had previously examined the relationship between physiological and perceptual parameters as a function of cadence as once performed for concentric cycling. Previous experiments on the effects of cadence in ECC cycling have shown that low cadence (e.g., 30 rpm) (23), high cadence (e.g., 75 rpm) (24), or both (25,26) resulted in greater cardiorespiratory responses compared with intermediate cadences close to 60 rpm and thus suggested a similar relationship to that found in concentric cycling. In the present study, cadence changes elicited different types of cardiorespiratory response depending on power output. Indeed, extreme cadences of 30 and 90 rpm elicited a greater cardiorespiratory response

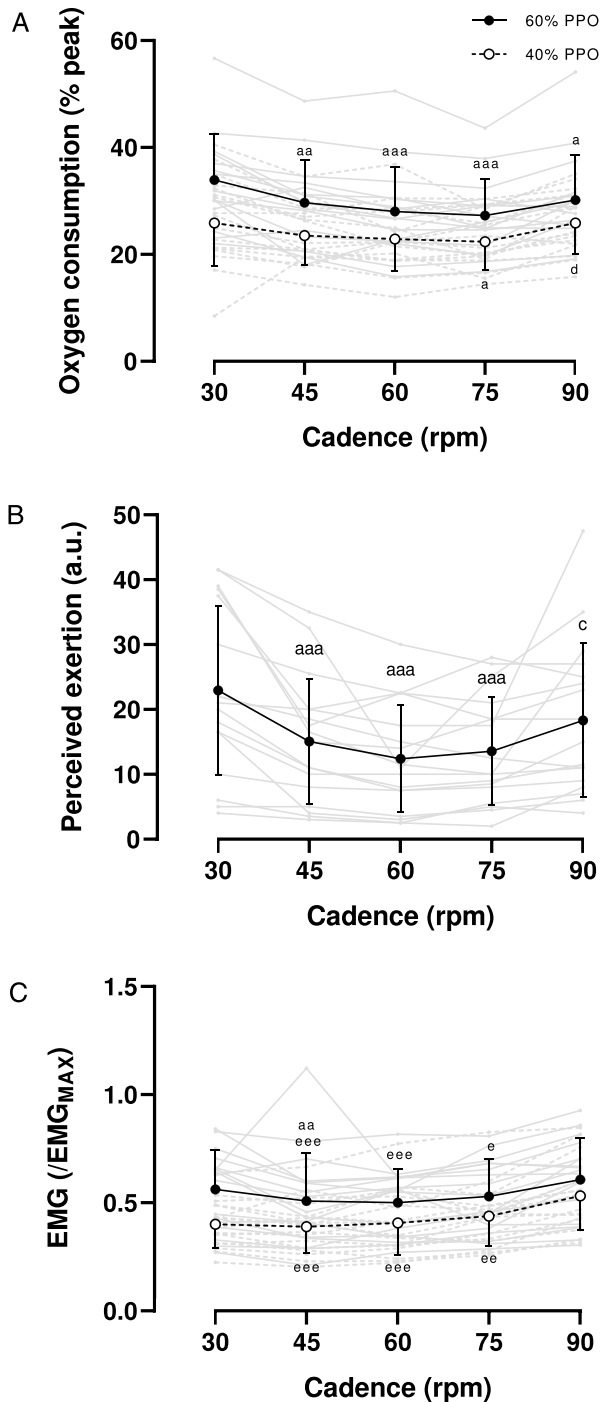


FIGURE 1—Mean (SD) of oxygen consumption (A), PE (B), and EMG (C) values for different cadences. Values of the two power outputs were averaged when two-way repeated-measures ANOVA did not show significant interaction between power and cadence. Gray lines represent individuals. a, different from the condition at 30 rpm; b, different from the condition at 45 rpm; c, different from the condition at 60 rpm; d, different from the condition at 75 rpm; e, different from the condition at 90 rpm. One sign, $P < 0.05$; two signs, $P < 0.01$; three signs, $P < 0.001$.

($25.9\% \pm 6.9\% \dot{V}O_{2peak}$) compared with intermediate cadences ($22.4\% \pm 5.2\% \dot{V}O_{2peak}$) at 40% PPO. At 60% PPO, only the low cadence of 30 rpm resulted in a greater response ($33.9\% \pm 8.6\% \dot{V}O_{2peak}$) than all other cadences ($28.8\% \pm 7.9\%$

$\dot{V}O_{2peak}$). Although there was no significant increase in cardio-respiratory response to high cadence at high intensity chosen in the present study, the responses to both power outputs fitted a curvilinear curve to changes in cadence.

Respiratory frequency has been previously studied, but no differences with cadence changes were found (24). The mentioned study focused on a range of cadences between 35 and 75 rpm, whereas our study evaluated the effect of cadence on respiratory frequency for a wider range of cadences (30 to 90 rpm). We found that the cadence of 90 rpm induced a higher respiratory frequency than lower cadences (except 75 rpm). A too narrow range of cadences may be insufficient to appreciate the effect of cadence on respiratory frequency. In addition, it has been proposed that the elevated respiratory frequency (with no change in minute ventilation) in ECC cycling could be explained by the need for the participants to hold the handlebar to stabilize their body posture, evidenced by an increase in upper limb muscle activation (32), which leads to limited inspiratory capacity (36). Preliminary experimentation in our laboratory suggested that a concomitant increase in biceps brachii muscle activation and respiratory frequency at both low (30 rpm) and high (90 rpm) cadences was an effect of an increase in the force applied on the pedal and lower limb movement, respectively, which result in the need to trunk stabilization. However, the present study did not corroborate the previous observations because minute ventilation and respiratory frequency followed a comparable pattern to cadence. At the same time, muscle activation of biceps brachii was much greater at low cadence (30 rpm) at 60% PPO only. Moreover, although biceps brachii activation increased with power output, it is possible that respiratory frequency did not increase as expected because the exercise intensity in the present study was not as high as in the experiment of Lechaue et al. (32) that compared concentric and ECC cycling performed at the same metabolic power.

Conventional concentric cycling studies have generally shown an increase in muscle activation at both low and high cadences. However, no study had yet investigated the interaction between muscle activation and cadences during ECC cycling. In the current study, we evaluated muscular activation of the lower limb and chose to average the muscular activation of the six muscles to represent an overall activation of the lower limbs. As with oxygen uptake, we found a curvilinear function with an optimal cadence for muscular activation. At 40% PPO, only high cadences generated greater muscular activation, whereas low and high cadences of 30 and 90 rpm caused an increase in muscular activation compared with the intermediate cadence at 60% PPO. The high cadence of 90 rpm seems to have an effect on the recruitment of additional motor units at high power output (60% PPO in the present study). Nevertheless, it might be insufficient to significantly increase muscular activation at lower intensities. Taking one at a time, the activation patterns differed between muscles. Knee extensors responded differently, given their anatomical role. Although the VL muscle showed greater activation at both low and high cadences, the RF muscle activation was greater at low cadences

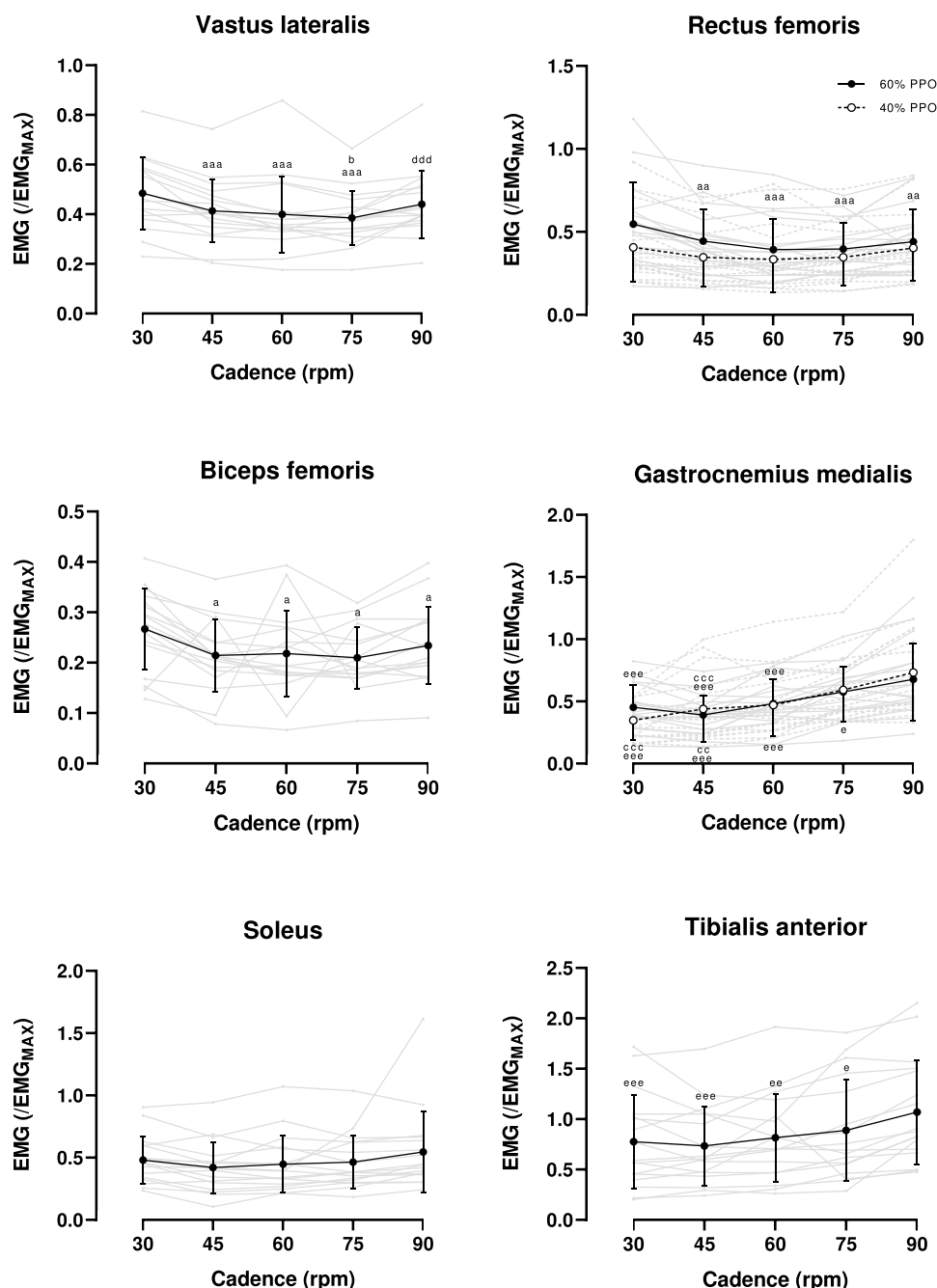


FIGURE 2—Mean (SD) of muscular activation of the different lower limb muscles. Values of the two power outputs were averaged when two-way repeated-measures ANOVA did not show significant interaction between power and cadence. *Gray lines* represent individuals. When one curve was represented, a *post hoc* analysis was performed for the main effect of cadence, and a sign was placed above the line. When two curves were presented, signs were placed below the line for 40% PPO and above for 60% PPO condition. a, different from the condition at 30 rpm; b, different from the condition at 45 rpm; c, different from the condition at 60 rpm; d, different from the condition at 75 rpm; e, different from the condition at 90 rpm. One sign, $P < 0.05$; two signs, $P < 0.01$; three signs, $P < 0.001$.

only. The monoarticular status of the VL muscle provides it a central role in force production during ECC cycling (37). By contrast, the biarticular RF muscle may be more importantly involved in the transition phases between leg extension and flexion (38). Knee flexors activation was also influenced by cadence. Although it has been proposed previously that the BF muscle did not have a crucial role in absorbing the power produced during ECC cycling (2), we found that this muscle was highly activated at low cadence. The plantar ankle flexors

(GM and SOL) were similarly activated when power output increased but their responses to changes in cadence were different. Indeed, although GM muscle activation increased at higher cadences, SOL muscle activation remained constant. This discrepancy could be explained by the biarticular anatomy of the GM muscle, which has a potential role in knee stabilization at high cadences. By contrast, the SOL muscle only plays a role in plantar flexion of the ankle. The dorsal ankle flexor muscle was highly activated at high power output and

Biceps brachii

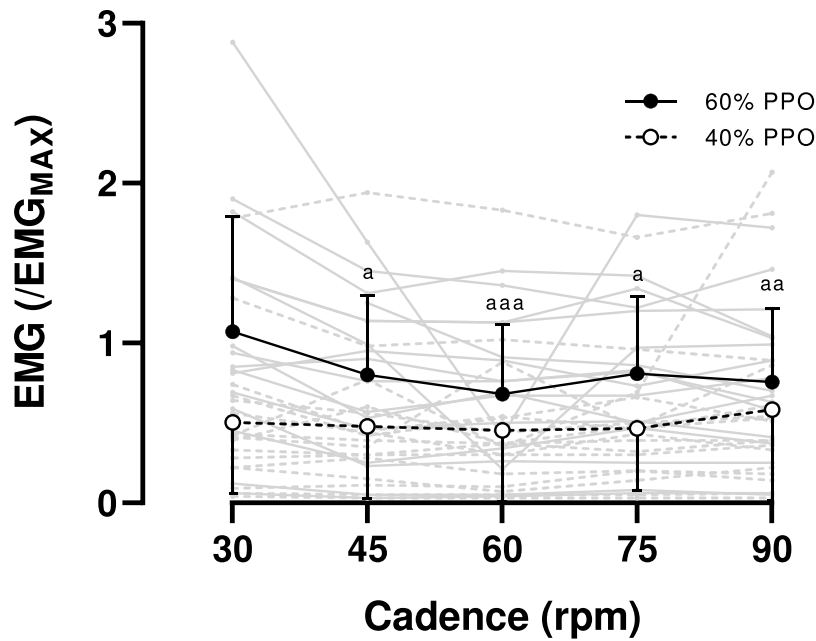


FIGURE 3—Mean (SD) of muscular activation of biceps brachii. Gray lines represent individuals. Statistical significance signs were placed below the line for 40% PPO and above for 60% PPO condition. a, different from the condition at 30 rpm. One sign, $P < 0.05$; two signs, $P < 0.01$; three signs, $P < 0.001$.

cadence. The TA muscle responded similarly to the GM muscle. Their anatomical role as antagonists in ankle stabilization could explain their similar coactivation pattern with cadence.

PE is used to set exercise intensity during rehabilitation programs (10); however, ECC cycling could result in greater muscle damage, soreness, and maximal voluntary torque decrease. If exercise intensity is set from conventional cycling, it is PE (39). In the present study, the effort was perceived as low, as illustrated by the mean value of 13/100 and 20/100 at 40% and 60% PPO, respectively, corresponding to a perception between

“weak” and “moderate.” Like metabolic demand, the perception of effort is mediated by and associated with the central motor command (related to voluntary muscle contraction) in the sense that the neural mechanism of corollary discharge generates the sensation of effort (40,41). The present experiment showed that extreme cadences induced greater perception of effort than intermediate cadences. Although the perception of effort reached 12/100 at 60 rpm (the most common cadence in ECC cycling studies), it increased to 23 and 18 for 30 and 90 rpm, respectively. Therefore, because a heightened perception of effort

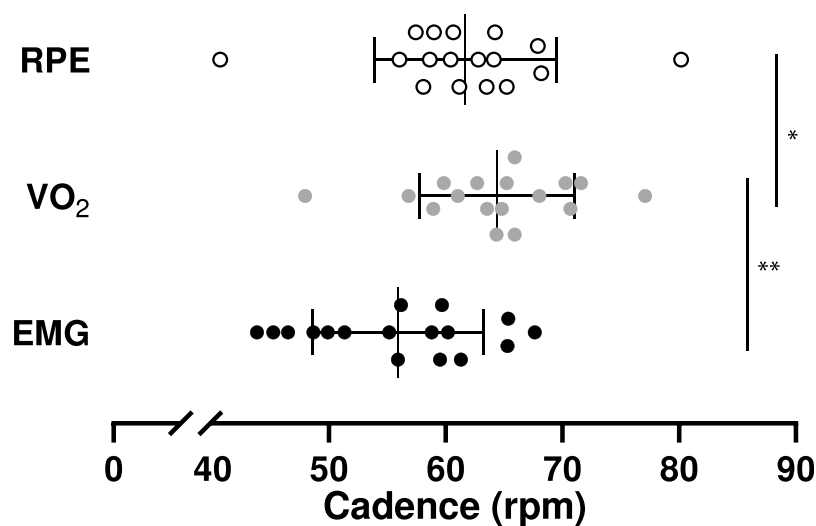


FIGURE 4—Estimated optimal cadences for PE (RPE), oxygen consumption ($\dot{V}O_2$), and global muscular activation (EMG). Values for both power outputs were averaged because two-way repeated-measures ANOVA did not show significant interaction between power and cadence. Statistical analysis revealed only a main parameters effect. Scatter lines represent the mean and SD. * $P < 0.05$. ** $P < 0.01$.

during a rehabilitation program may hinder engagement in a regular physical activity program (42), attention to patients' perception of exertion during a new exercise is necessary.

As previously mentioned, physiological and perceptual responses followed U-shaped curves as a function of change in cadence with high confidence (mean r value = 0.84). The estimated optimal cadence of oxygen consumption response was greater (64.5 ± 7.9 rpm) than that of muscle activation (55.9 ± 9.3 rpm) and PE (61.7 ± 9.4 rpm) responses. By contrast, it was not different between the last two parameters. This difference between optimal cadences has been previously discussed for conventional cyclist exercise (18). Although the perception of effort for concentric cycling seems to represent a trade-off between muscle tension and oxygen consumption, the present data suggest that the minimization of effort does not coincide with the minimum $\dot{V}O_2$ response but is closer to the minimization of lower limb muscle action. However, from a practical point of view, these differences between the three estimated optimal cadences were minimal (<10 rpm). The present results did not confirm our hypothesis from concentric cycling (16) regarding the effect of power output on optimal cadences. An upward shift in optimal cadence with power in concentric cycling has generally been attributed to the force-velocity properties of muscle fiber types (19) and the proportion of fiber types recruited (43). However, in ECC cycling, the range of power output available was wider compared with concentric cycling (44); then it is possible that

intensities in the present study were too close to induce a shift through the recruitment of high-threshold motor units.

CONCLUSIONS

In conclusion, during ECC cycling, physiological and perceptual responses to changes in cadence followed a curvilinear function, leading to extreme cadences of 30 and 90 rpm eliciting greater physiological and perceptual responses than intermediate cadence. The limb muscles' activation followed distinct patterns according to their anatomical role. Estimated optimal cadences that minimize oxygen consumption, muscular activation, and perception of effort were close to 60 rpm. Professionals should be careful in choosing of cadence when offering ECC cycling to patients with cardiorespiratory limitations/diseases because the effect of the session may be too high than expected. Indeed, our study evaluated low power outputs in a healthy population, and further research should be conducted in a specific population. Finally, because the cadence alters physiological and perceptual responses, fatigability should be considered after prolonged ECC cycling exercise performance at different cadences.

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